

Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

Izma Alhazmi Herdian^{1, 2, a)}, Sulthan Naufal Affan^{1, b)}, and Estiyanti Ekawati^{1, 2, c)}

¹Engineering Physics Department, Institut Teknologi Bandung, Bandung 40132, Indonesia

²Center for Instrumentation Technology and Automation, Institut Teknologi Bandung, Bandung 40132, Indonesia

^{a)}Corresponding author: izmaherdian@gmail.com

^{b)}sulthannaufal96@gmail.com ^{c)}esti@itb.ac.id

Abstract. Surveying a non-convex site with a quadrotor swarm must remain safe and complete when the wind is strong, obstacles are unknown in advance, and agents fail in flight. We present a three-layer architecture for this task. The high level partitions the region by polygon-clipped centroidal Voronoi tessellation, plans boundary-following boustrophedon sweeps and reallocates the unswept lanes of failed agents. The mid level maps obstacles with on-board LiDAR and filters every velocity command through a control barrier function quadratic program whose class- $\mathcal{K}$ function follows from the tilt limit and dead time of the vehicle. The low level is a cascaded PID with gains synthesised from the LQR or the H_∞ Riccati equation. Seven quadrotors were flown in ROS 2/Gazebo over one convex and three non-convex regions under Gauss–Markov turbulence ($\sigma = 2.5$ m/s) over a 3.0 m/s mean wind, nine unmapped static and two moving obstacles, and two scheduled agent losses. Seven of eight missions finished with 96.9–99.0% coverage despite losing 28.6% of the fleet. PID- H_∞ completed all four regions, lowered the RMS lateral sweep error by 16–39%, mostly by rejecting the mean wind, and finished 11–15% sooner than PID-LQR where both finished, at 3.9–5.6 times its torque effort. One PID-LQR mission was lost when an agent touched an obstacle, and both controllers grazed the same obstacle in one region. Each configuration was flown once, so these are single-run observations.

INTRODUCTION

Multi-rotor swarms shorten aerial surveys, but real sites are rarely rectangles: they have re-entrant corners, gusty wind, obstacles unknown before take-off, and vehicles that fail. Centroidal Voronoi tessellation (CVT) with Lloyd’s iteration [1] distributes agents over a region and has been combined with obstacles and actuator faults [2], safety constraints [3], non-convex domains [4] and agent dropout [5]; boustrophedon sweeps remain a standard way to cover a cell [6, 7], including cooperative UAV surveys [8] with re-planning after failures [9]. Control barrier functions (CBF) enforce safety as a quadratic program (QP) on the input [10, 11], with high-order [12], multi-robot [13] and quadrotor forms for moving obstacles [14] and wind gusts [15]; LQR and H_∞ synthesis remain common for multirotor control under wind [16, 17], and swarm fault tolerance is an active topic [18]. Hierarchical quadcopter swarms with a reactive potential-field planner above cascaded PID loops fly formations through clutter and gusts, yet purely reactive planners remain prone to U-shaped traps [19].

What is less reported is how these pieces behave *together* on a physically simulated vehicle, where velocity commands are not realised instantly and wind drift costs coverage. This paper contributes (i) an architecture running polygon-aware CVT partitioning, boundary-following sweeps and lane-level fault reallocation above a LiDAR-mapped CBF-QP whose class- $\mathcal{K}$ function is derived from the closed-loop model of the vehicle; (ii) a comparison of PID gains from the LQR and H_∞ Riccati equations, on the linear model and inside the swarm; and (iii) a data-traceable evaluation of eight missions under combined hazards, including their failures. Every reported result is computed by an analysis script from the telemetry.

METHODOLOGY

System Architecture

Figure 1 shows the three layers, defined per agent: a cell, a lane queue, a safety filter using neighbours within 6 m and obstacles within 10 m, and a flight controller. The upper layers of all agents run in one 20 Hz ROS 2 node [20],

sharing the coverage grid, obstacle map and failure notices without delay, and each vehicle runs its own 250 Hz low level; distributing the upper layers is future work.

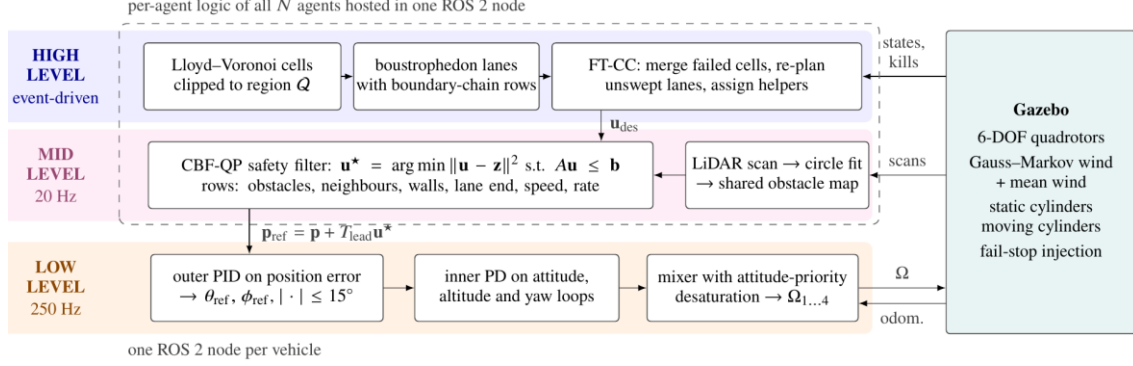


FIGURE 1. Three-layer architecture: per-agent layers, with the high and mid levels of all agents hosted in one ROS 2 node and one low-level controller per vehicle.

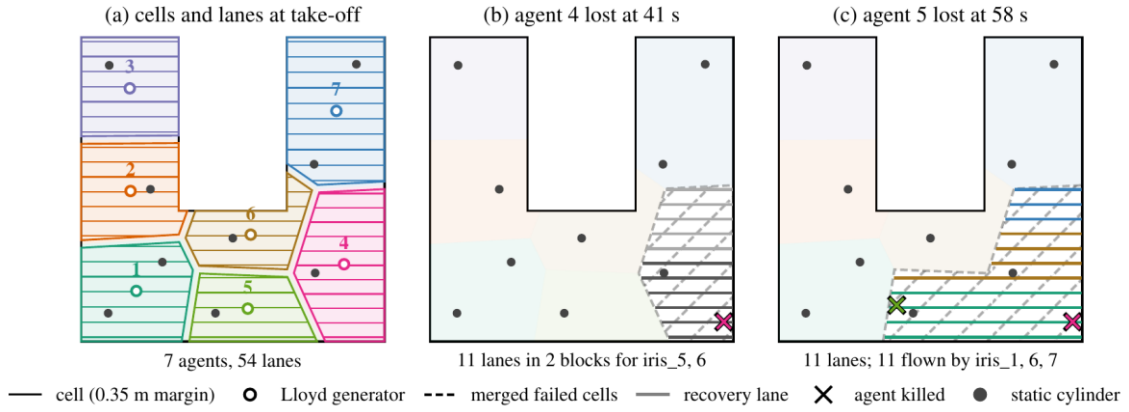


FIGURE 2. High-level planning in the $\text{PID-}H_\infty$ U-shape mission, reproduced with the planner code (centroids within 0.6 cm of the log, identical lane counts). (a) Cells and their 54 lanes; first and last lanes follow the cell boundary. (b) Agent 4 is lost; its cell yields 11 lanes for agents 5, 6. (c) Helper 5 is lost too; the merged cells yield 11 lanes, coloured by the agent that flew them.

High Level: Partition, Sweep and Fault-Tolerant Cooperative Coverage

Given the region polygon Q and generators $\mathbf{g}_i$, the cell of agent i is

$$\mathcal{P}_i = \{q \in Q: \|q - \mathbf{g}_i\| \leq \|q - \mathbf{g}_j\|, \forall j \neq i\}, \quad (1)$$

computed by clipping bisector half-planes and intersecting the result with Q . Lloyd's iteration $\mathbf{g}_i \leftarrow \text{centroid}(\mathcal{P}_i)$ (25 iterations from deterministic k-means++ seeds) minimises $\sum_i \int_{\mathcal{P}_i} \|q - \mathbf{g}_i\|^2 dq$ [1], and agents are assigned to cells

by the Hungarian algorithm. Each cell is shrunk by 0.35 m and swept by east-west lanes at most 1.45 m apart; its first and last lanes follow the cell boundary, so the wedges that a fixed-pitch sweep leaves at slanted Voronoi edges are flown too (Fig. 2a). With a sensing radius $r_s = 0.95$ m the swath is 1.90 m (overlap of at least 0.45 m). Coverage is scored on a 0.30 m grid: a grid cell inside Q counts once its centre has been within r_s of a live, sweeping agent flying above 0.8 m; grid cells under static obstacles are excluded.

When agent k fails, Fault-Tolerant Cooperative Coverage (FT-CC) keeps the recovery lanes that k still carried, erases the coverage recorded inside $\mathcal{P}_k$ (the data of the lost vehicle are treated as lost), merges all failed cells, re-plans lanes over their union and drops lanes that are already at least 80% covered. The helpers are the neighbours of the failed cells closest to the recovery centroid (one for a single loss with at most five lanes, otherwise two, and three after a second loss). They receive contiguous serpentine blocks, which they fly at once if their own cell is finished and append to their queue otherwise (Fig. 2b,c).

Mid Level: Obstacle Mapping and CBF-QP Safety Filter

Obstacle map. Each 360-ray scan is cleaned of returns within 0.55 m of other agents and 0.80 m of the arena boundary, clustered, and fitted with circles; detections form one swarm-wide track list, a track is confirmed after 12 hits and rejected if it moves, and only confirmed tracks enter the QP, so the swarm starts with an empty map. The planner deliberately ignores obstacles, leaving avoidance to the safety filter. The two moving obstacles are taken from simulator odometry, i.e. assumed perfectly tracked.

Actuator-consistent barrier. With the reference rule $\mathbf{p}_{\text{ref}} = \mathbf{p} + T_{\text{lead}}\mathbf{u}$, the outer PID makes the commanded-velocity response first order with an acceleration limit, $\dot{v} = \text{sat}_{\pm a_{\text{max}}}(k_v(u - v))$, where

$$k_v = gK_d, \quad a_{\text{max}} = g \tan \theta_{\text{max}}, \quad T_{\text{lead}} = \frac{K_d - k_{\text{ff}}}{K_p}, \quad v_c = \frac{a_{\text{max}}}{k_v}, \quad (2)$$

with $\theta_{\text{max}} = 15^\circ$ and $k_{\text{ff}} = 0.15$ rad s/m. With the LQR outer-loop gains (used for both controllers) $k_v = 3.867 \text{ s}^{-1}$, $a_{\text{max}} = 2.631 \text{ m/s}^2$, $T_{\text{lead}} = 0.323 \text{ s}$ and $v_c = 0.680 \text{ m/s}$. The stopping distance from speed $s > v_c$ is $D(s) = (s^2 + v_c^2)/2a$; requiring that the distance travelled during a lumped dead time $T_d = 0.40 \text{ s}$ plus $D(s)$ fit inside the clearance h gives the admissible approach speed

$$\varphi(h) = \begin{cases} \max(0, -aT_d + \sqrt{\max(0, a^2T_d^2 + 2ah - v_c^2)}), & h \geq 0, \\ \gamma_r h, & h < 0, \end{cases} \quad (3)$$

which is exactly zero for $h \leq v_c^2/2a$, unlike a linear γh (Fig. 3a); the branch $h < 0$ ($\gamma_r = 3.5 \text{ s}^{-1}$) demands an outward velocity once the set has been left. The budget is $a = \max(0.15a_{\text{max}}, \mu a_{\text{max}} - |a_{\text{obs}}| - |a_{\text{wind}}|)$ with $\mu = 0.75$ (0.85 for walls) and a wind allowance of 0.5 m/s^2 .

Quadratic program. For an obstacle or neighbour at $\mathbf{o}$ with $\hat{\mathbf{n}} = (\mathbf{p} - \mathbf{o})/\|\mathbf{p} - \mathbf{o}\|$, the condition $\dot{h} \geq -\varphi(h)$ is the linear row

$$-\hat{\mathbf{n}}^T \mathbf{u} \leq \lambda \varphi(h) - \hat{\mathbf{n}}^T \mathbf{v}_o, \quad h = \|\mathbf{p} - \mathbf{o}\| - (\rho + r_d + \delta), \quad (4)$$

with drone radius $r_d = 0.22 \text{ m}$, obstacle radius ρ (fitted radius + 0.05 m for mapped cylinders, 0.50 m for moving ones), margins $\delta = 0.35 \text{ m}$ (static) and 0.75 m (moving), and $\lambda = 1$ for obstacles. Neighbour rows are reciprocal ($\lambda_{ij} + \lambda_{ji} = 1$, weighted by mission state) at a hard radius of 0.70 m and a soft radius of 1.20 m ; walls and the end of the current lane give half-plane rows, and inscribed polygons bound the speed (3 m/s) and the per-tick change of $\mathbf{u}$. Each agent then solves

$$\mathbf{u}^* = \underset{\mathbf{u} \in \mathbb{R}^2}{\text{argmin}} \|\mathbf{u} - \mathbf{z}\|^2 \quad \text{s.t.} \quad A\mathbf{u} \leq \mathbf{b}, \quad \mathbf{z} = \frac{\mathbf{u}_{\text{des}} + t + w\mathbf{u}_{\text{prev}}}{1 + w}, \quad (5)$$

exactly, by enumerating the unconstrained point, the projections onto every row and the pairwise vertices (Fig. 3b). A tangential bias $\mathbf{t}$, applied within a 60° cone of a head-on approach, breaks symmetric deadlocks and enters only the objective; $w = 0.25$ smooths successive commands. If the QP is infeasible, soft rows are released (tier 1), then the violation of the hard rows is minimised (tier 2), and a scaled previous command is the last resort (tier 3).

Low Level: LQR and H_∞ Gain Synthesis

Each axis is modelled about hover as a double integrator $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ (e.g. $B = [0, g]^T$ from pitch to x , $B = [0, 1/I_y]^T$ from torque to pitch) and augmented with an integrator on the input, $\mathbf{x}_a = [\mathbf{x}^T, \mathbf{u}]^T$. The LQR design solves the algebraic Riccati equation (ARE) for $K_a = R^{-1}B_a^T P$. The H_∞ design treats a disturbance entering through the input channel, $B_w = [B^T, 0]^T$, and solves

$$A_a^T P + P A_a + Q_a - P(B_a R^{-1} B_a^T - \gamma^{-2} B_w B_w^T) P = 0, \quad (6)$$

rejecting solutions for which $A_a - B_a K_a$ is not Hurwitz [16], with $\gamma = 1.3 \gamma_{\text{min}}$ and γ_{min} found by bisection.

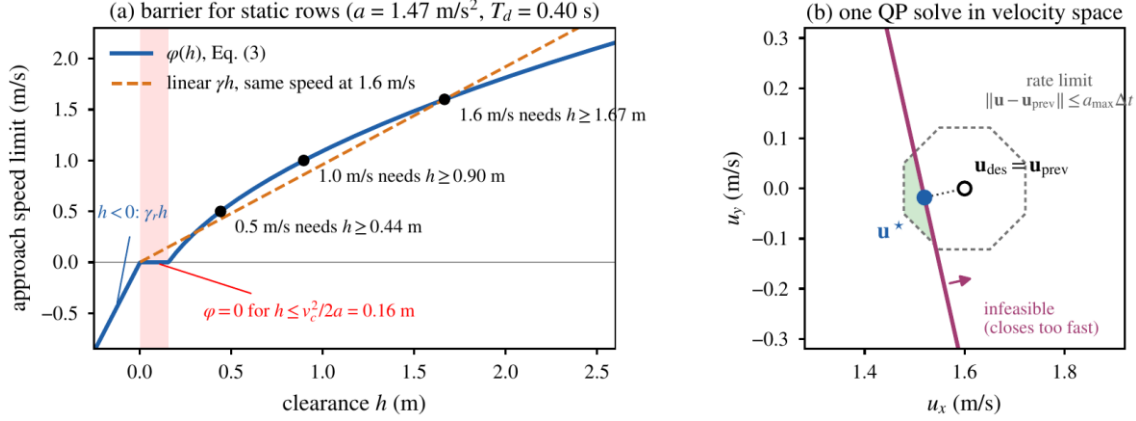


FIGURE 3. Mid-level safety filter with mission parameters. (a) $\varphi(h)$ of Eq. (3) for static rows: zero below $h = 0.16$ m, $h \geq 1.67$ m needed at 1.6 m/s; a linear γh permits approach where φ demands a stop. (b) One QP solve by the flight code for an illustrative case (cylinder 2.51 m ahead): the closing speed drops from 1.56 to 1.48 m/s.

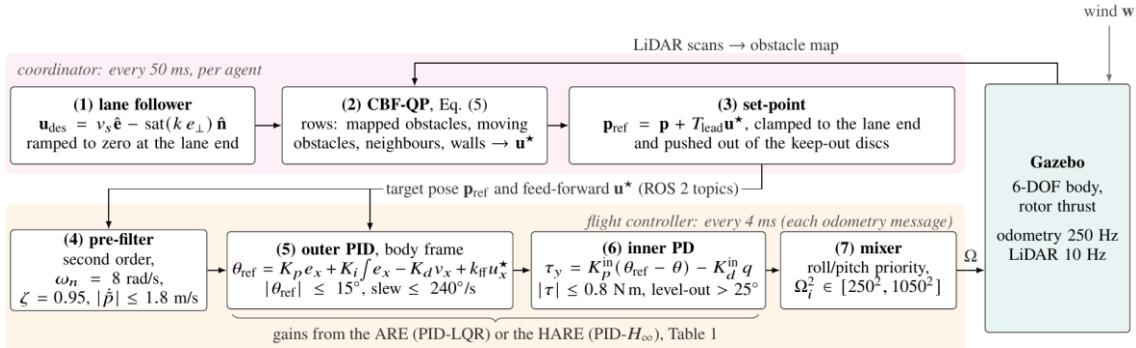


FIGURE 4. Signal path of one control step. The coordinator (1)–(3) publishes a set-point and a velocity feed-forward every 50 ms; the flight controller (4)–(7) runs on each 4 ms odometry message. The controllers differ only in the gains of (5)–(6) and in the position integrator (PID- H_∞ : always active, clipped to ± 0.6 ; PID-LQR: active within 0.25 m, clipped to ± 0.4).

The state-feedback gain is mapped to PID form through the pseudo-inverse of $\Gamma = [C^T, (CA)^T, (CA^2)^T; 0, (CB)^T, (CAB)^T]$; the synthesised inner integral gain is not used. Figure 4 shows where the gains enter the signal path of one control step, and Table 1 lists weights, attenuation levels and gains.

Simulation Setup

Seven 3DR Iris quadrotors ($m = 1.50$ kg, $I_{xx} = I_{yy} = 0.0291$, $I_{zz} = 0.0552$ kg m², arm 0.22 m, $k_f = 8.549 \times 10^{-6}$ N s², rotor speed ≤ 1100 rad/s) are simulated in Gazebo [21] with ROS 2. Controllers read ground-truth odometry; no state estimator is in the loop. Missions start from a launch pad south of the region, cruise at 2.0 m and sweep at 1.6 m/s. Four regions inside a 28×28 m box are used (Fig. 5): *rect* (784 m²), *L-shape* (588 m²), *U-shape* (624 m², 10 m wide notch) and *plus* (460 m², 10 m wide arms), each with nine vertical cylinders ($r = 0.40$ m, 4 m tall) placed inside the region. All missions use the same combined-hazard scheme:

- *Wind.* A first-order Gauss–Markov process per horizontal axis ($\sigma = 2.5$ m/s, correlation time 0.5 s, seed 42), a surrogate for the Dryden and von Kármán models used in quadrotor studies [22], plus a constant -2 m/s mean on both axes from $t = 5$ s, uniform in space and applied through the Gazebo wind-effects plugin (vertical component scaled by 0.4). Recorded mean $(-2.34, -1.95)$ m/s, standard deviation $(2.45, 2.51)$ m/s, horizontal peaks 12.0 m/s.
- *Moving obstacles.* Two kinematic cylinders ($r = 0.45$ m) oscillate along the region diagonals with amplitude 10 m (12 m, offset by 1 m, in *plus*) at 0.15 and 0.11 rad/s, i.e. peak speeds of 1.6–2.5 m/s.
- *Agent loss.* When coverage first reaches 18% and 32%, one pre-selected agent is killed (rotor speeds set to zero, fail-stop) and the coordinator is notified at once. Victims were chosen to own comparable cells next to the launch pad (Table 2).

TABLE 1. Gain synthesis (outer x/y and inner roll/pitch loops) and linear analysis of the outer loop with the inner loop closed. IGL_{∞} : peak gain from a disturbance acceleration to position; PM/GM: phase and gain margins of the outer loop.

Design	Loop	Q, R	$\gamma_{\min} \rightarrow \gamma$	K_p	K_i	K_d	$IGL_{\infty} (s^2)$	PM	GM
PID-LQR	outer	diag(0.08, 0.35), 1	—	0.757	0.283	0.394	0.158	37.4°	11.5 dB
	inner	diag(6, 0.6), 0.05	—	4.868	—	0.534			
PID- H_{∞}	outer	diag(0.10, 0.40), 1	2.54→3.30	2.705	1.092	1.107			
	inner	diag(4, 0.4), 0.06	2.54→3.31	11.604	—	1.189	0.040	25.2°	9.0 dB

TABLE 2. Mission results. T : mission time; lat.: RMS lateral sweep error; $\bar{v}$: mean sweep speed; alt.: RMS altitude error; tilt: 95th percentile while sweeping, and maximum; effort: mean $\tau_x^2 + \tau_y^2$ (N^2m^2); d_{obs} , d_{v2v} : minimum surface distance to a static cylinder and inter-agent distance; tier: solves needing tier 1–2. Killed: rect 4, 7 (H_{∞}), 7 (LQR); L 4, 7; U 4, 5; plus 1, 7.

Region	Controller	Status	Cov. (%)	T (s)	lat. (cm)	$\bar{v}$ (m/s)	alt. (cm)	tilt ₉₅ (°)	tilt _{max} (°)	effort	d_{obs} (m)	d_{v2v} (m)	tier (%)
rect	PID- H_{∞}	done	98.2	260	7.2	1.20	2.5	14.7	21.1	0.054	0.49	1.71	2.9
	PID-LQR	abort@78.8	n/a	n/a	11.7	1.12	7.3	14.7	155.9	0.024	0.06	1.52	n/a
L-shape	PID- H_{∞}	done	98.4	204	8.4	1.19	2.5	14.5	20.8	0.058	0.57	1.32	3.8
	PID-LQR	done	98.5	236	12.2	1.11	2.6	14.5	26.3	0.010	0.29	0.79	4.2
U-shape	PID- H_{∞}	done	98.5	205	8.3	1.21	2.6	14.8	20.7	0.044	0.63	1.77	4.1
	PID-LQR	done	96.9	240	11.4	1.07	2.5	14.2	16.6	0.011	0.42	1.73	4.1
plus	PID- H_{∞}	done	98.4	198	9.7	1.15	5.4	14.7	82.5	0.053	0.24	1.24	4.1
	PID-LQR	done	99.0	222	11.5	1.02	2.6	14.4	40.1	0.014	0.24	1.12	4.9

Both controllers share the planner, filter and plant model (Eq. (2) with LQR gains). Each region–controller pair was flown once ($N=1$) and recorded on video; the two *plus* missions also include three transit guards (stall detector, speed reduction above 16° tilt, 20 s start-wait timeout). Metrics use the 20 Hz controller logs without the first 20 s and without rows after a kill. Lateral error is taken during steady east–west sweeps (heading stable for 1.5 s, speed above 0.6 m/s), torque effort is the mean of $\tau_x^2 + \tau_y^2$, and obstacle distance is measured from the drone centre to the cylinder surface; the airframe reaches 0.235 m (body) to 0.348 m (rotor tips), so smaller distances mean certain or possible contact. We also assume a lightweight rotor guard, as on collision-tolerant multirotors [23], so that grazing contacts load the guard rather than the propellers; the guard is not simulated and changes no reported number.

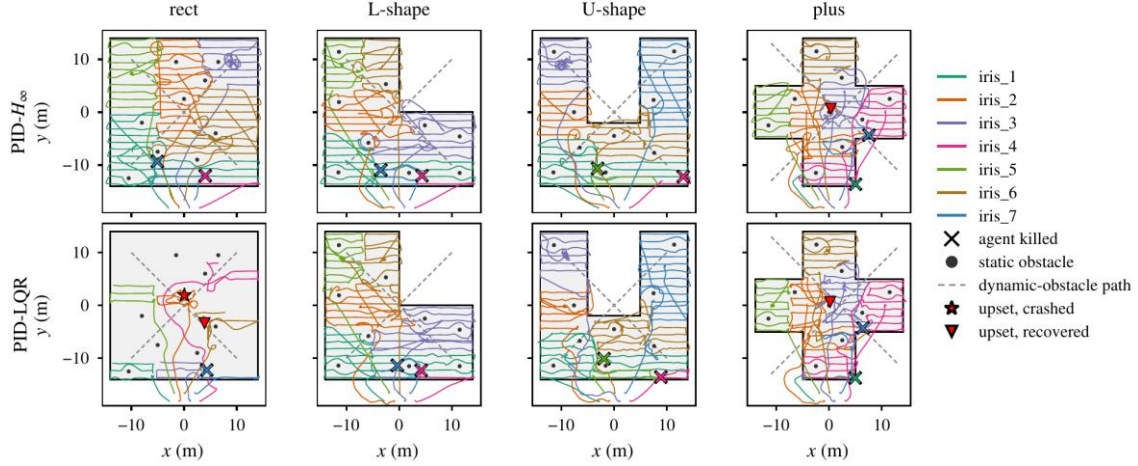


FIGURE 5. Flown trajectories of live agents above 0.5 m, including transit from the launch pad ($y \approx -18$ m).

RESULTS AND DISCUSSION

Mission Completion and Fault Recovery

Table 2 and Fig. 5 summarise the eight missions. Seven finished with all survivors back at their cell centroids and 96.9–99.0% coverage, despite losing two of seven agents at 18% and 32% coverage. Every obstacle map held all nine cylinders (centre and radius within the 0.1 m log resolution) plus at most two unmatched tracks, and no completed mission needed tier 3. Each kill causes a small drop, as the failed cell’s coverage is erased, before the slope resumes (Fig. 6); Fig. 7 shows the final maps. The PID-LQR U-shape mission ended at 96.9% with thin uncovered strips along cell boundaries in the left arm, nearly absent for PID- H_{∞} ; their cause was not isolated. The PID-LQR rect mission was aborted at 78.8 s.

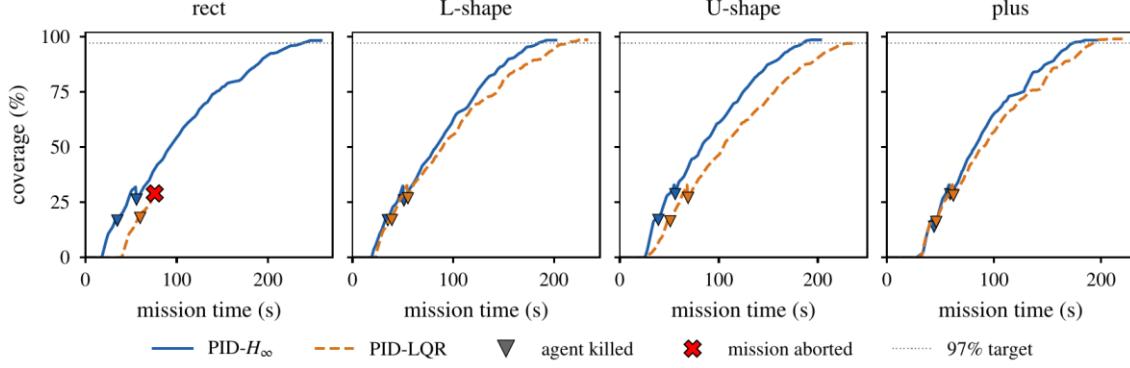


FIGURE 6. Coverage against mission time (coordinator clock). Triangles: agent kills, after which the failed cell is re-swept.

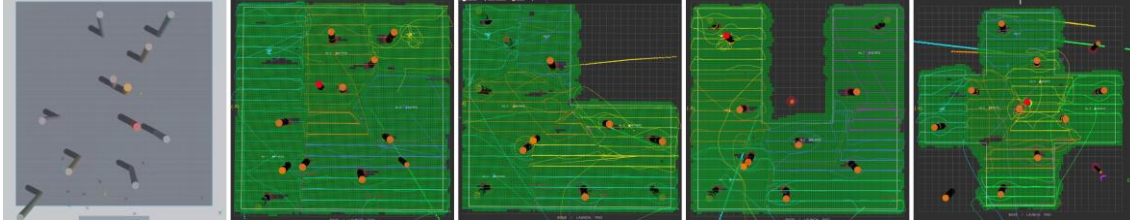


FIGURE 7. (a) Gazebo, rect scenario (screenshot); (b)–(e) final RViz coverage maps (green) of the $PID-H_\infty$ missions, from the recordings.

Tracking under Turbulence: LQR versus H_∞ Gains

The two designs differ clearly in how tightly they hold a lane (Fig. 8a): the median lateral error was 2.7–3.8 cm with $PID-H_\infty$ and 8.5–8.8 cm with $PID-LQR$, and the RMS error fell from 11.4–12.2 cm to 7.2–9.7 cm (16–39%) in all four regions. Most of the difference is a mean offset: $PID-LQR$ flew –5.7 to –7.0 cm downwind (towards $-y$), $PID-H_\infty$ within 0.7–1.6 cm of the lane, with closer standard deviations (7.0–9.6 vs 9.4–10.8 cm). The H_∞ outer integral gain is 3.9 times larger, and the $PID-LQR$ node freezes its position integrator beyond 0.25 m of error, which happened in 36–45% of its sweep samples. Mean sweep speeds of 1.15–1.21 against 1.02–1.12 m/s (command 1.6 m/s) account for the 11–15% shorter missions; the 95th-percentile tilt was essentially equal, as was altitude error outside the two missions with upsets. In the linear analysis (Table 1) the H_∞ gains lower the worst-case disturbance-to-position gain by 4.0 times and the position deviation under the Gauss–Markov input by 3.7 times (Fig. 8c).

The price is a higher effort: $PID-H_\infty$ used 3.9–5.6 times the torque effort of $PID-LQR$ (Fig. 8b), with motor saturation below 0.2% of samples, and its outer loop has a higher crossover (9.08 vs 4.09 rad/s) and smaller margins. The measured reduction is far below the linear factor because tilt limit, pre-filter, QP rate limits and wind force model are not in the linear model. As Q , R differ and the integrator gating belongs to the $PID-LQR$ node, this compares two complete gain designs and does not isolate the effect of γ from that of higher loop gain.

Safety: Clearance, Contact and Upset Events

No two agents came closer than 0.79 m ($PID-LQR$, L-shape), above the 0.70 m hard radius. The minimum surface distance to static cylinders was 0.49, 0.57 and 0.63 m under $PID-H_\infty$ and 0.06, 0.29 and 0.42 m under $PID-LQR$ in rect, L- and U-shape; in *plus* both passed within 0.24 m of the central cylinder, inside the rotor envelope, where the assumed guard takes the contact. The upsets (Fig. 9) share a pattern: each happened while the agent stepped to its next lane beside a cylinder, after 1–4 s of tier-2 solves bound by that cylinder. In the $PID-LQR$ rect mission agent 2 came within 0.06 m of the cylinder at (0,2.5) (body contact), tilted to 155.9° and fell. In *plus*, agent 2 of both missions grazed the central cylinder, tilting to 40.1° ($PID-LQR$) and 82.5° ($PID-H_\infty$, dropping to 0.64 m and recovering within about 3 s). With one run per case these events cannot be attributed to the controller; they are better explained by obstacle-blind lanes ending inside a cylinder’s margin, a barrier built on a 0.22 m disc, and a fixed 0.5 m/s² wind allowance that ignores gusts of up to 12.0 m/s. A 50.8° upset ($PID-LQR$, rect, agent 6, 24.4 s) before sweeping, 0.55 m from the nearest cylinder, was recovered.

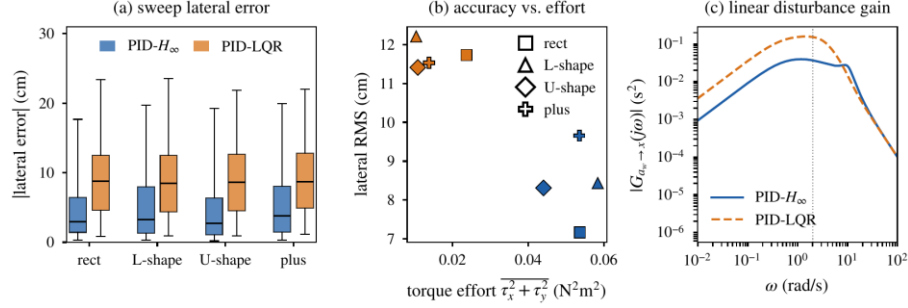


FIGURE 8. (a) Absolute lateral error in steady east–west sweeps (box: quartiles, whiskers: 5th–95th percentiles). (b) RMS lateral error against torque effort. (c) Disturbance-to-position gain of the linearised outer loop; dotted: wind-model corner frequency.

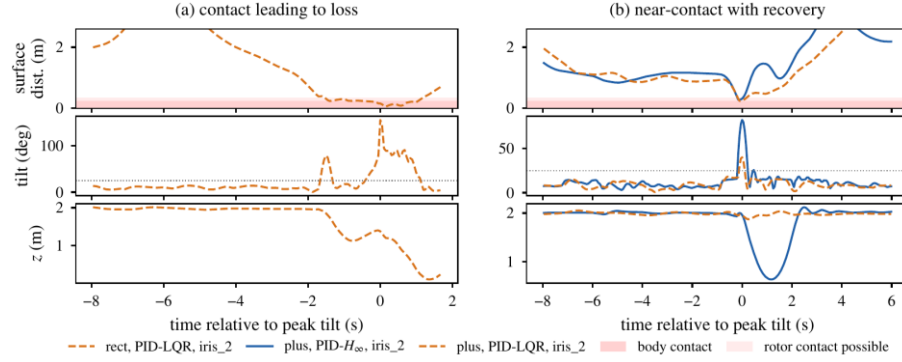


FIGURE 9. Upsets aligned at peak tilt: surface distance to the nearest static cylinder (dark: body contact; light: rotor contact possible), tilt (dotted: 25° level-out) and altitude. (a) PID-LQR, rect, agent 2: contact and loss. (b) *plus*, agent 2, both controllers: graze and recovery.

In all 8 missions some tier-2 solves bound by a moving-obstacle row reported a negative minimum barrier value (down to -0.68 m): a margin was entered, but the log keeps only the minimum over all rows and no moving-cylinder positions, so a contact can be neither confirmed nor excluded; under the guard assumption it would load the guard. Since the guard enlarges the envelope, a guard-sized barrier radius, obstacle-aware lane ends, a gust-dependent acceleration budget and logging of the moving obstacles are the next steps.

Limitations

The results rest on one run per configuration (fixed wind seed), perfect state feedback and moving-obstacle tracking, instantaneous failure detection, geometric coverage, assumed but unsimulated rotor guards and centralised upper layers; no ablation or re-implemented baseline.

CONCLUSION

A three-layer architecture of polygon-clipped Voronoi partitioning, boundary-following sweeps, lane-level fault reallocation, a LiDAR-mapped actuator-consistent CBF-QP and Riccati-synthesised PID control completed seven of eight simulated missions with 96.9–99.0% coverage under turbulence, unmapped and moving obstacles, and the loss of two of seven quadrotors. H_∞ gains held the lanes 16–39% more tightly and, where both finished, shortened missions by 11–15% relative to LQR gains, at several times the torque effort and with smaller margins; both designs grazed static obstacles, and one contact ended an LQR mission. Next steps are repeated runs over wind seeds, a guard-sized, wind-aware barrier, distributed upper layers and flight tests.

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